Lab # 10

More About Functions in C++

Objective

To understand advanced concepts of functions in C++ such as:

- Function with parameters
- Function with return value
- Function calling with arrays
- Reusability of code using functions

Apparatus / Software Required

- Computer
- C++ Compiler (Code::Blocks / Dev C++ / Online Compiler)

Theory

Functions in C++ are used to divide a program into small parts. Each function performs a specific task.

Key Points

- Functions can take **parameters (inputs)**
- Functions can **return values**
- Functions help in **code reuse**
- They make programs **simple and organized**

Example:

Program:

Function with parameter and return value

```cpp
#include <iostream>
using namespace std;

// Function to add two numbers
// yeh function do numbers ko add karta hai aur result return karta hai
int add(int a, int b) {
    return a + b;   // a aur b ko add karke wapas return kar raha hai
}

int main() {
    int x, y, sum;   // x aur y user input ke liye, sum result store karega

    cout << "Enter two numbers: ";
    cin >> x >> y;   // user se do numbers input liye ja rahe hain

    sum = add(x, y);   // function call karke sum calculate kiya ja raha hai

    cout << "Sum = " << sum << endl;   // final result screen par print ho raha hai

    return 0;}   // program successfully end hota hai
}
```

Working

- The function add() takes two numbers as input
- It returns their sum
- In main(), user inputs two values
- Function is called and result is displayed

Output Example

Enter two numbers: 5 7

Sum = 12

Conclusion

In this lab, we learned how functions can take inputs and return values. Functions make programs more efficient, reusable, and easy to understand.

Lab 10 Tasks

1. Function with Parameters

▪ Declare a function called ”multiply” that takes two integer parameters:”num1”and”num2”.

▪ Inside the function, calculate the product of the two numbers.

▪ Display the result within the function.

▪ Call the function from the main program, passing two numbers, to perform the multiplication

Code:

```
#include <iostream>
using namespace std;

// Function declaration
// yahan hum function ka naam aur parameters define kar rahe hain
void multiply(int num1, int num2);

int main() {
    int a, b;   // user se input lene ke liye variables

    cout << "Enter two numbers: ";
    cin >> a >> b;   // user se do numbers liye ja rahe hain

    // Function call
    // yahan function ko call karke multiplication karwaya ja raha hai
    multiply(a, b);

    return 0;   // program successfully end hota hai
}

// Function definition
// yahan function ka actual kaam likha gaya hai
void multiply(int num1, int num2) {
    int product = num1 * num2;   // dono numbers ko multiply kiya ja raha hai
    cout << "Product = " << product << endl;   // result screen par print ho raha hai
}
```

2: Function with Return Value

▪ Declare a function called ”getSquare” that takes an integer parameter: ”number”.

▪ Inside the function, calculate the square of the number.

▪ Return the result.

▪ Call the function from the main program, passing a number, and display the square.

Code:

```
#include <iostream>
using namespace std;

// Function declaration
// yahan hum function ka naam aur return type (int) define kar rahe hain
int getSquare(int number);

int main() {
    int num, result;   // num user input ke liye, result square store karega

    cout << "Enter a number: ";
    cin >> num;   // user se ek number input liya ja raha hai

    // Function call
    // yahan function ko call karke square hasil kiya ja raha hai
    result = getSquare(num);

    cout << "Square = " << result << endl;   // result screen par print ho raha hai

    return 0;   // program successfully end hota hai
}

// Function definition
// yahan function number ka square calculate karta hai
int getSquare(int number) {
    return number * number;   // number ko usi se multiply karke return karta hai
}
```